U.S. Department of Transportation
ITS Joint Program Office

Crash Scene Reconstruction Using Unmanned Aircraft System Imagery
North Carolina DOT



Source: Getty Images, Inc.

Challenges

The North Carolina Department of Transportation (NCDOT) and the North Carolina State Highway Patrol (NCSHP) sought to evaluate whether unmanned aircraft systems (UAS) and photogrammetry processing techniques could improve the speed, safety, and efficiency of crash scene documentation and reconstruction activities. The agencies were particularly interested in reducing roadway closure durations, improving responder safety, accelerating data collection, and supporting more efficient postcrash investigations.

Legacy crash reconstruction and investigation methods are time consuming and can prolong roadway closures during major investigations. Two existing methods of investigating collisions, total station and laser scanning, can increase exposure time of first responders to traffic. The existing methods also limit the speed at which traffic operations can be restored.

Agency Background

NCDOT led this crash reconstruction research effort through its Division of Aviation and UAS Program Office in partnership with the NCSHP Collision Reconstruction Unit (CRU). NCDOT's Division of Aviation served as the primary technical and operational lead for the UAS activities. The Division of Aviation's UAS flight team was responsible for conducting UAS flight operations, aerial imagery collection, and photogrammetric processing associated with the research demonstration. The Division of Aviation describes its role as ensuring safe and responsible UAS operations in North Carolina. NCDOT provides research and education to demonstrate how beneficial and safe the technology can be. The NCSHP CRU provided crash investigation expertise and led the comparison with legacy reconstruction methods. The CRU conducts investigations that lead to identification, apprehension, and prosecution of offenders. The unit frequently provides evidential support in high-profile collision cases.

At a Glance

AGENCY: **North Carolina Department of Transportation**

USE CASE AREA: **Transportation safety – crash reconstruction**

MATURITY LEVEL: **Exploratory research/field demonstration**

METHOD: **Unmanned aircraft system (UAS)-based photogrammetry and 3D reconstruction**

DATA SOURCE: **UAS aerial imagery, video, and ground control point measurements**

KEY OUTCOME: **Reduced crash scene mapping time from 1 hour and 51 minutes to 25 minutes (~77% faster), with measurement accuracy within 0.03 feet**

Method Used

This exploratory deployment evaluated UAS and photogrammetry processing software to support crash scene reconstruction and investigation activities. The effort focused on determining whether aerial imagery and three-dimensional (3D) mapping workflows could improve the speed, safety, and efficiency of crash scene documentation compared with legacy methods of investigating collisions: total station and laser scanning. The generated outputs allowed investigators to view and measure crash scenes from multiple perspectives. These outputs supported accurate measurement of roadway and vehicle geometry and documented skid marks and impact locations with high spatial accuracy.

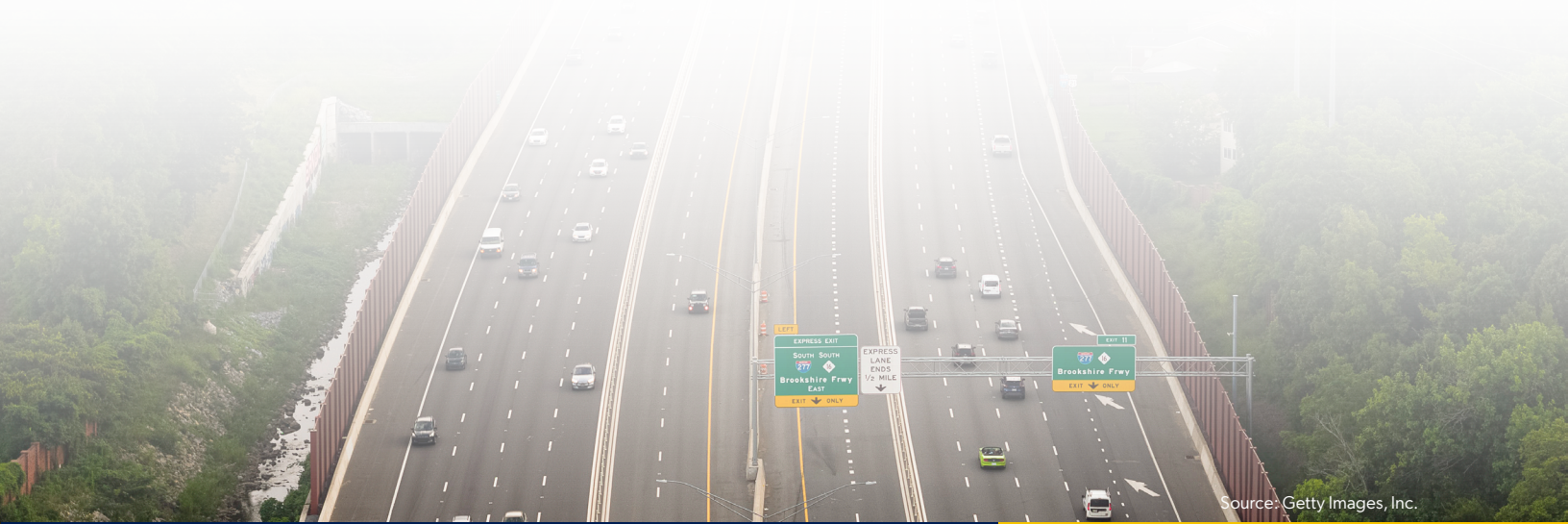
Although national research has noted that machine learning (ML) and image-processing techniques may eventually support automated feature extraction from UAS-generated datasets, the North Carolina deployment itself primarily evaluated photogrammetry-based reconstruction workflows rather than operational ML-enabled crash analysis. Investigators retained full responsibility for interpreting evidence, validating measurements, determining contributing factors, and reaching investigative conclusions. As a result, the technology functioned as a digital documentation and mapping capability rather than an automated decisionmaking or enforcement system.

Development Timeline

NCDOT and the NCSHP conducted the crash reconstruction research effort in 2017 through a joint evaluation of UAS for collision scene documentation and reconstruction. Follow-on national transportation safety research later cited the North Carolina effort as an example of how UAS-generated imagery and related analytical techniques could support future crash reconstruction and safety analysis applications.

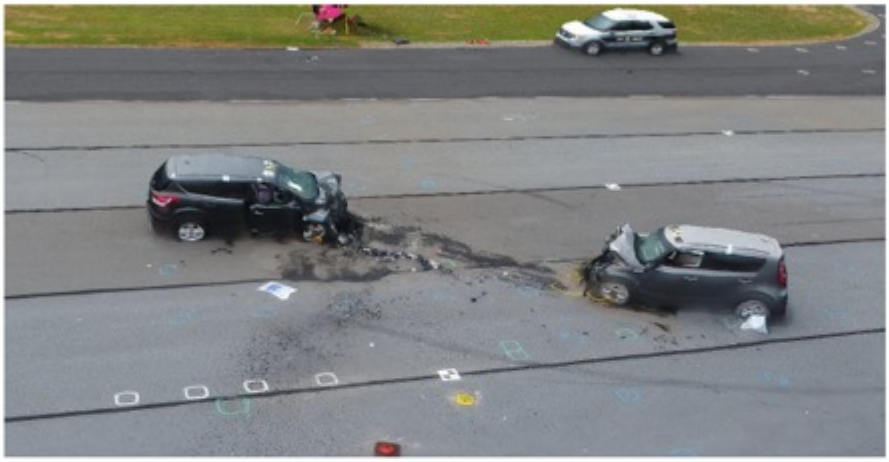
Development Process

NCDOT and the NCSHP developed the exploratory crash reconstruction workflow through a structured field demonstration comparing UAS-based photogrammetry against legacy laser scanning methods. The effort focused on evaluating whether UAS and photogrammetry processing software could support crash scene documentation while reducing roadway closure durations and exposure risks for investigators. During the demonstration, NCDOT's Division of Aviation used three different UAS platforms to capture aerial imagery of a simulated head-on collision scene. The flight team collected aerial imagery and video using the three UAS platforms and processed the imagery using photogrammetry processing software to generate orthophotos (i.e., aerial photographs geometrically corrected to have a uniform scale and aligned with real-world map coordinates), dense point clouds, and 3D scene representations.



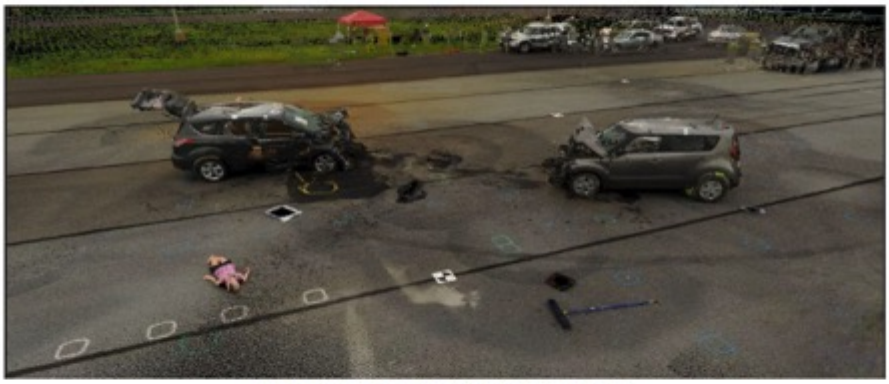
Source: Getty Images, Inc.

Figure 1 compares the point cloud renderings from photogrammetry processing software and the terrestrial laser scanner. The two photos show similar crash scenes of two vehicles with similar nearby skid marks and pavement markings. The research team took point clouds from both systems for rendering into a 3D model of the crash scene. The files from each system were similar in size (i.e., approximately 12 million points each). The team noted that renderings from each system produced great detail, making it easier to recognize skid marks and accurately read targets from medium flying heights. The learning objective of the figure is to highlight that the photogrammetry software should allow investigators to create an accurate and time-saving product.



Source: NCDOT.

(a) Rendering from photogrammetry processing software.



Source: NCDOT.

(b) Rendering from terrestrial laser scanner.

Figure 1. Illustrations. Point cloud rendering comparison.

The photogrammetry workflow included image alignment, point-cloud generation, mesh construction, texture generation, and 3D-model export for reconstruction review. To evaluate accuracy, the agencies established six ground control point (GCP) targets and compared measured locations with the actual ground locations. The study found that the UAS-generated products produced reconstruction-quality measurements comparable to legacy methods while substantially reducing field collection time. NCSHP's CRU mapped evidence of the same crash scene using a laser scanner, allowing a data comparison between the UAS-based and legacy reconstruction approaches. The agencies then evaluated differences in mapping time, processing workflows, and data accuracy to support the crash scene reconstruction.

Figure 2 shows the software measuring the distance between the two vehicles. The calculation originates from the six GCP targets established before the acquisition of imagery. The photogrammetry software processed the data to create an orthomosaic (i.e., a single, continuous image created by combining overlapping aerial photographs that have been geometrically corrected to accurately represent the ground surface at a consistent scale). The software's image quality tool found the image quality to be a value of 0.87, which was well above the minimum value of 0.60 recommended by the software. Then, using independent manual measuring methods, the research team found the overall accuracy of software-measured locations to be within 0.03 feet. The team also noted that the photogrammetry software facilitated accurate measurement of skid marks and of the relative distance between vehicles after impact.



Source: NCDOT.

Figure 2. Screenshot. Distance calculation using photogrammetry processing software.

Resources

The study used commercially available UAS, photogrammetry processing software, laser scanning equipment, and supporting field survey resources to evaluate crash scene reconstruction workflows. Technical resources included three different UAS platforms operated by NCDOT's UAS flight team and aircraft equipped with high-resolution cameras. Staffing requirements primarily involved UAS pilots, crash reconstruction investigators, survey personnel, and technical staff supporting reviews of imagery processing and reconstruction. The demonstration included a two-person UAS flight team from NCDOT's Division of Aviation and investigators from the NCSHP CRU. Field surveying activities, which the agency used to set up GCP targets for evaluating measurement accuracy, also supported the demonstration.



Source: Getty Images, Inc.

Outputs, Outcomes, and Impacts

The study produced several digital mapping and scene documentation outputs used to evaluate the feasibility of photogrammetry-supported crash reconstruction workflows. These outputs included georeferenced orthophotos, dense point clouds, 3D scene models, and geometric measurements derived from aerial imagery collected during the demonstration. The generated products supported crash scene visualization, measurement, and documentation activities while allowing comparison against legacy terrestrial laser scanning methods.

The exploratory UAS photogrammetry demonstration showed that crash scene documentation could be completed substantially faster than legacy laser scanning methods while still producing reconstruction-quality geometric data. During the study, the UAS flight team completed aerial mapping operations in 25 minutes, while the terrestrial laser scanning workflow required approximately 1 hour and 51 minutes to map the same scene.

The report noted that the ability to clear collisions and restore traffic rapidly is of significant interest to NCDOT, because faster clearance "reduces exposure time of first responders to traffic and helps preserve the safety of personnel at the scene. The faster a highway can be reopened to traffic, the better NCDOT can maintain the mobility of roads and the commerce that passes through them."

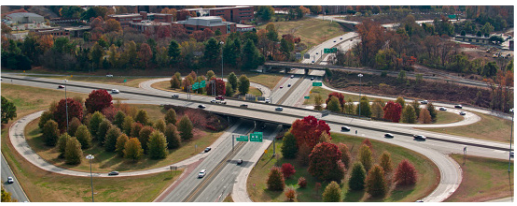
The study also demonstrated the feasibility of integrating commercially available UAS platforms and photogrammetry software into crash scene documentation workflows.

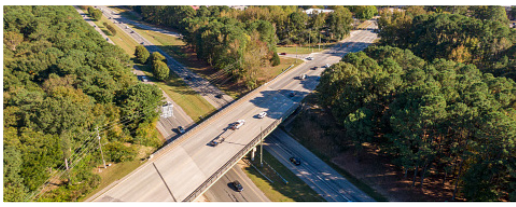
Development Challenges

Operational limitations associated with UAS deployment presented challenges. The study noted that the demonstration occurred during daylight and favorable weather conditions, and the agencies identified the need for further research involving nighttime operations, rain, snow, and other adverse environmental conditions that could affect imagery quality or flight operations.



Source: Getty Images, Inc.





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Lessons Learned

- **Early field demonstrations supported understanding of operational feasibility.** NCDOT and the NCSHP used a controlled crash scene demonstration to evaluate how UAS-supported photogrammetry compared with legacy laser scanning methods. The side-by-side comparison helped investigators assess differences in speed, visualization capability, and reconstruction accuracy.
- **Rapid aerial imagery collection reduced roadway closure time.** The study demonstrated that UAS-supported mapping could document large crash scenes in 25 minutes, substantially faster than the laser scanning workflow evaluated during the demonstration. Faster field collection has the potential to improve roadway reopening time and reduce exposure risks for investigators and motorists.
- **Photogrammetry products provided reconstruction-quality geometric documentation.** Orthophotos, dense point clouds, and 3D scene models generated from aerial imagery provide sufficiently accurate measurements for reconstruction review and scene documentation activities. Validation using surveyed GCPs helped confirm the accuracy of the reconstructed products.
- **Environmental conditions are an important operational consideration.** The agencies noted that more research would be needed to evaluate nighttime operations, adverse weather conditions, and other field constraints that could affect imagery quality or UAS operations.

Transferability

The experience showed that UAS-supported photogrammetry can offer a practical and efficient method for documenting crash scenes while preserving investigator-led reconstruction review. The study showed that commercially available UAS platforms and photogrammetry software could generate accurate digital scene documentation products while potentially reducing roadway closure durations and improving responder safety.

References

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North Carolina Department of Transportation, Division of Aviation. 2017. *Collision Scene Reconstruction & Investigation Using Unmanned Aircraft Systems*. Raleigh, NC: North Carolina Department of Transportation Division of Aviation. https://www.ncdot.gov/divisions/aviation/Documents/ncshp-uas-mapping-study.pdf.



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